FUNDING PROPOSAL · UNIVERSITI PENDIDIKAN SULTAN IDRIS

Pengetahuan **Suluh** Budiman
Est. 1922 · Tanjung Malim

SULUH

Southeast Asia's first university-built sovereign educational AI — trained, governed, and served entirely inside UPSI.

A 35-billion-parameter model bound to four institutional harnesses under one PDPA-compliant governance layer. Malaysian student data never leaves Malaysian soil.

THE ASK	TIMELINE	BREAKEVEN	LICENSING
RM 450,000	12 months	Month 14	RM 0

v3.3 · 21 July 2026
UPSI Research Management

UPSI has been first before. The torch is already in our motto — *Pengetahuan Suluh Budiman.*

1922

First teachers' college

Sultan Idris Training College opens in Tanjung Malim — the cradle of Malay education and intellect.

1997

First education university

SITC becomes UPSI — Malaysia's first university dedicated to education itself.

2006

Digital pioneer

MyGuru makes UPSI an early national adopter of open-source learning platforms.

2026

First sovereign educational AI

SULUH — to our knowledge, the first higher-education-focused sovereign AI built and owned by a university in Southeast Asia.

Suluh (Malay): a torch — the light that guides. The system is named by the university's own founding motto.

For RM 450K over 12 months, UPSI owns a sovereign educational AI that reaches cost-parity with the cloud by Month 14 — and never sends a single student record offshore.

SITUATION 95% of students already use AI. UPSI's grading, advising, and analytics are heading the same way — and every leading option runs in a foreign cloud.

COMPLICATION Cloud AI puts student PII under foreign jurisdiction just as PDPA penalties rose to **RM 1 million with personal liability for officers** — and bills compound at RM 15–25K monthly, forever.

RESOLUTION Build it on-premise once. A 35B sovereign model, four harnesses, and IsRAG provenance on every answer — **RM 0 software licensing**, breakeven in 14 months.

WHAT RM 450K BUYS

Sovereign compute cluster	RM 200K
Engineering team, 12 mo	RM 250K
Annual OPEX thereafter	RM 30K
36-MO SAVING VS CLOUD	**RM 240–600K**

Sources: SULUH v3.3 economic analysis · HEPI Student Generative AI Survey 2026 · PDPA (Amendment) Act 2024 · model tags verified live 20 Jul 2026

Six questions, one decision.

Why

Students moved first, attackers moved next, the law moved last — the cloud is now the wrong foundation.

What

One system, two fundable halves — and a first-of-its-kind position in the region.

Who

Four harnesses owned by the UPSI units that live the problem — and what UPSI gains.

Where

Five desktop nodes and 10,000 laptops, all inside the firewall.

When

Three tracks in parallel; sovereign serving live from Month 2.

How

Novel research incl. IsRAG, governance in code, and economics that break even.

The last question — How — resolves into the decision on the final slide.

PART ONE · THE IMPERATIVE

Why.

Three forces converged in twenty-four months: students adopted AI faster than any technology in university history, universities became the world's most-attacked sector, and Malaysian law grew teeth. The foreign cloud is now the wrong foundation.

The question is no longer whether UPSI adopts AI — students already did. The question is whose AI, under whose law.

95% of students now use AI in some form — up from 66% just two years ago

94% use generative AI on **assessed work** — grading integrity is already an AI problem

38% of institutions provide AI tools. The other 57 points of usage are **ungoverned shadow AI**

Every day that UPSI has no institutional AI, students and staff fill the gap themselves — pasting assignments, drafts, student records, and research data into **free consumer chatbots hosted overseas**, under terms of service no one has read.

That data trains someone else's model, sits under someone else's jurisdiction, and is governed by no UPSI policy. **Doing nothing is not a neutral option — it is an unmanaged data-export programme.**

> SULUH replaces shadow AI with a governed alternative that is **better** for students — bilingual, curriculum-aware, and free — not just a policy that bans what we cannot see.

Source: HEPI/Kortext Student Generative AI Survey 2026 (n=1,054 undergraduates) · 2024→2026 trend: 66% → 92% → 95% any-AI use

Universities are now the most-attacked institutions on earth — and the biggest breaches came through third-party vendors, not campus firewalls.

4,388
cyberattacks per education organisation **every week** — the highest of any sector (Q2 2025)

3.9M
education records exposed in confirmed 2025 ransomware attacks — up 27% year-on-year

39%
of identity-based attacks Microsoft observed targeted research and academic institutions

CASE · UNIVERSITY OF PHOENIX

3.5M people exposed via one vendor zero-day

Attackers exploited Oracle's E-Business Suite weeks before the vendor even knew. Three months of dwell time before detection. Names, birthdates, ID numbers — gone.

CASE · COLUMBIA UNIVERSITY

870K records, 460 GB, two months undetected

One phished phone call. Attackers roamed the network from May to June 2025, taking academic histories, financial aid files, and health information.

CASE · POWERSCHOOL

62M students exposed by one ed-tech vendor

A single compromised support credential — no MFA — reached the records of 18,000 schools. The largest student-data breach in history was a **supply-chain** failure.

Sources: Check Point via DeepStrike (Q2 2025) · Comparitech Education Ransomware Roundup 2025 · Microsoft Digital Defense Report 2025 · public breach disclosures 2025 · CISA: education is "target rich, cyber poor"

Since 2025, mishandling personal data is a RM 1 million offence — and liability reaches named officers, personally.

RM 1M

maximum fine per offence — up from RM 300K — plus up to **3 years' imprisonment**

Personal

directors, managers, and officers are **personally liable** unless they prove due diligence

72 hrs

mandatory breach notification to the Commissioner — failure alone costs up to **RM 250K**

DPO

a registered Data Protection Officer is **mandatory since June 2025** at UPSI's data scale

Cross-border

Offshore transfers are only lawful to jurisdictions with **substantially similar protection** — a test each foreign AI vendor must pass, and UPSI must evidence, for every data flow.

Every cloud AI query that carries a student record is a cross-border transfer UPSI must justify, a vendor contract UPSI must audit, and a breach surface UPSI cannot see into **— yet UPSI keeps 100% of the liability**.

Sovereignty inverts the equation: when data never leaves campus, the hardest compliance questions **never arise**. SULUH doesn't manage this risk. It removes it.

Source: Personal Data Protection (Amendment) Act 2024, in force Jan–Jun 2025 · JPDP Guidelines Feb 2025 · avg. Malaysian breach cost est. RM 3.2M incl. fines, legal & churn (Simply Data 2025)

Every cloud option forces student data offshore — SULUH makes the question structurally impossible to ask.

Grading, advising, and staff analytics all run on real student and employee records. Sending them to a foreign API means UPSI can no longer guarantee where that data lives, who can subpoena it, or how it is retained.

Sovereignty is not a feature of SULUH — it is the design constraint. **No raw student PII ever leaves UPSI.** Cloud teachers touch only anonymised curriculum data.

Jurisdiction
Foreign cloud = foreign legal reach over Malaysian student records — including subpoena powers UPSI cannot contest.

Consent & purpose
PDPA requires granular, purpose-bound consent — unenforceable across a vendor boundary you cannot audit.

Breach accountability
72-hour notification and audit trails are UPSI's obligation — even when the breach is the vendor's, as in every 2025 case study.

PDPA = Personal Data Protection Act 2010, as amended 2024 · governance architecture detailed in Part Six

Cloud AI is a bill that never stops; a sovereign build is a cost that ends.

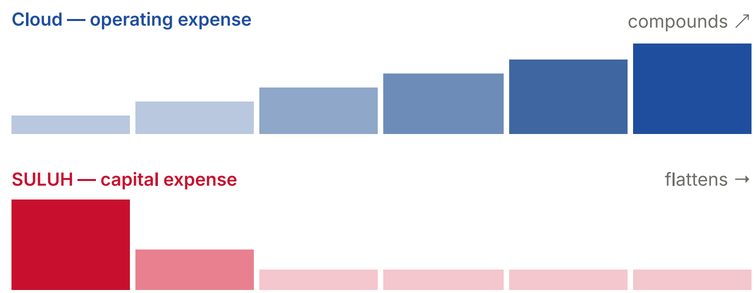
RM 15–25K per month, every month, for as long as UPSI uses AI

Cloud pricing scales with usage — the more UPSI adopts AI, the faster the meter runs. There is no asset at the end, no licence owned, no capability retained.

RM 0 software licensing — the entire stack is open source

SULUH spends CAPEX once and RM 30K/year to run. After the build phase the curve flattens; the cloud's keeps climbing.

TWO COST STRUCTURES



Breakeven at Month 14. Full quantified curve in Part Six.

The sovereign-AI wave is cresting now — and no university in the region has staked the higher-education claim. The window belongs to the first mover.

THE MOMENT

- **100+ countries** committed to AI sovereignty in the Bangkok Declaration (Feb 2026) — sovereignty is now the global default, not the exception.
- Malaysia's national push — **ILMU, NAIO, MyDIGITAL** — is funded and moving, but it targets nation-scale and industry. **Education is unclaimed.**
- Analysts warn that institutions which don't build become **"permanent renters"** of foreign AI — paying forever, owning nothing.

WHAT THE FIRST MOVER WINS

The standard: UPSI's governance gates and golden set become the national reference for educational AI.

The partnerships: ministries, MQA, and 20 public universities need exactly this — the first builder hosts the conversation.

The talent: the researchers and students who build sovereign AI come to — and stay at — the place that builds it.

The cost of one year of waiting: RM 180–300K in cloud spend, twelve more months of shadow-AI exposure — and the first-mover title passes to another campus.

Sources: Bangkok Declaration (Feb 2026) · Digital in Asia, Sovereign AI in Asia 2026 · Fulcrum/ISEAS 2026 · regional landscape detailed in Part Two

PART TWO · THE SYSTEM

What.

SULUH is one system with two fundable halves: a sovereign model, and the institutional harnesses that put it to work — occupying a position no one in the region holds.

Two projects, funded separately, operationally inseparable — that separation is what makes each one fundable.

SULUH AI

R&D · this ask

The sovereign model — a 35B-parameter educational AI trained, adapted, and served entirely on-premise.

RM 450K
200K hardware + 250K manpower

Grant
peer-reviewed research novelty

SULUH Ecosystem

Deployment-ready

The institutional harness layer — four production applications consuming SULUH AI as their inference backend.

Separate
operational IT budget

Infra
code already exists

> A research novelty needs peer review and reproducibility; institutional infrastructure needs operational budgets. Splitting them lets each be funded through the right channel — this deck is the bridge.

A 35B mixture-of-experts delivers 70B-class quality while fitting our hardware six times over.

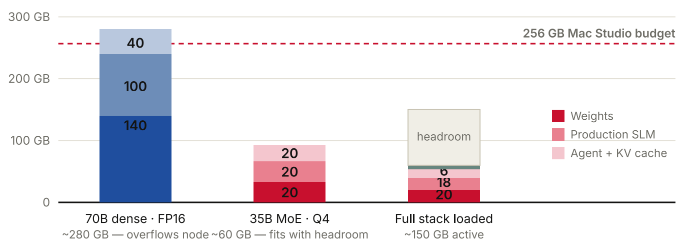


Figure — memory footprint. The full stack (teacher + production + agent + embedding) serves in ~150 GB of 896 GB aggregate.

A 70B dense model needs ~280 GB once weights and KV cache are counted — it overflows a single node. A 35B MoE activates just **3B parameters per token**, matching quality at a fraction of the memory.

6×
headroom on the cluster

Apache 2.0
Qwen3.6-35B-A3B, no licence fee

Full 100B pre-training is a Phase-2 vision contingent on national compute — **upside, not a Year-1 dependency**.

Nations are spending tens of millions on sovereign AI. Nobody has claimed higher education — SULUH takes that seat for 0.2% of Singapore's budget.

PROGRAMME	BACKER	INVESTMENT	FOCUS
NMLP / SEA-LION Singapore	IMDA · AISG · A*STAR	**S$70M** ≈ RM 230M	National, general-purpose regional LLM
ILMU Malaysia	YTL Corp × Universiti Malaya	**Corporate** · 500MW DC	National Bahasa Malaysia model, from scratch
Sahabat-AI Indonesia	GoTo Group	**Corporate**	Consumer & government services ecosystem
SULUH Malaysia · UPSI	A public university	**RM 450K**	**Education-domain, university-owned, on-campus** — first of its kind in SEA

National programmes prove the thesis and supply the upside — SULUH is designed to plug into NAIO/MyDIGITAL compute for its Phase-2 100B vision.

The education university building educational AI is not a coincidence of budget — it is **exactly the institution that should define this category**.

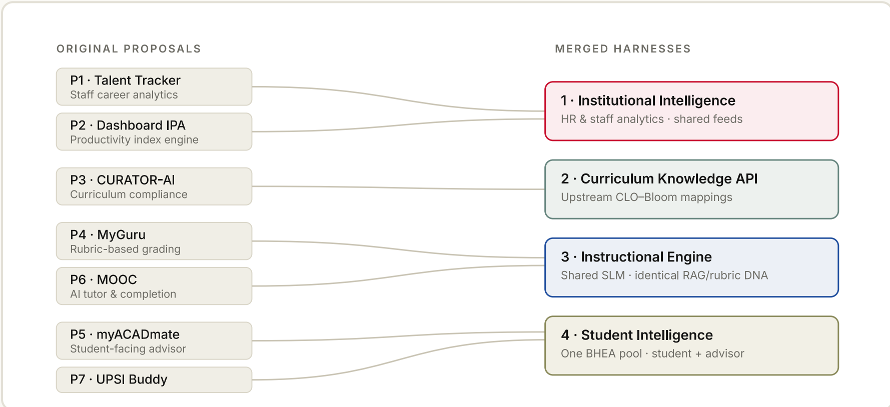
Sources: IMDA/AISG NMLP announcements · Fulcrum (ISEAS) 2026 · Digital in Asia 2026 · "first of its kind": no university-owned, higher-ed-domain sovereign LLM programme identified in SEA as of Jul 2026

PART THREE · OWNERSHIP

Who.

Seven scattered proposals collapse into four harnesses — each owned by the UPSI unit that already lives the problem. And the whole university rises with them.

Seven proposals consolidate into four harnesses on one governed data fabric — no duplicate backends.



1 **Institutional Intelligence** — BSM
Dr. Ahmad Amri · Dr. Nurul Hila

2 **Curriculum Knowledge API** — PPA
Dr. Amelia

3 **Instructional Engine** — ICT
Dr. Mohd Muslim · Dr. Ahmad Wiraputra

4 **Student Intelligence** — BHEA
Dr. Vasanthan · Pn. Nurul Ashikin · Jason Wong

P1/P2 share HR data · P4/P6 share grading & tutoring · P5/P7 share the student pool · P3 becomes an upstream API

SULUH is an institutional elevator: it lifts UPSI's research standing, its classrooms, its talent, and its national position — simultaneously.

Research standing

NeurIPS · ACL · Nature MI

Five planned publications on a **10,000-device evaluation testbed few labs on earth can replicate**. Citations, rankings, and grant capital compound from here.

Teaching & learning

Every student, every lecturer

Rubric-faithful grading in seconds, a bilingual tutor on every laptop, advising that sees the whole student record — **without a single record leaving campus**.

Talent & capability

Owned, not rented

An in-house team that can train, align, and govern frontier-class models — and graduates who learned on a sovereign stack. **This capability cannot be bought later at any price.**

National position

The education university

When ministries, MQA, and 20 public universities look for educational AI done right, they will look at **the university whose motto carries the torch**. That seat is taken by whoever builds first.

PART FOUR · THE FOOTPRINT

Where.

Everything runs on five desktop nodes and 10,000 campus laptops — all inside the UPSI firewall. No data centre required.

Five desktop nodes aggregate 896 GB; 10,000 laptops train the edge — and only adapter deltas ever move.

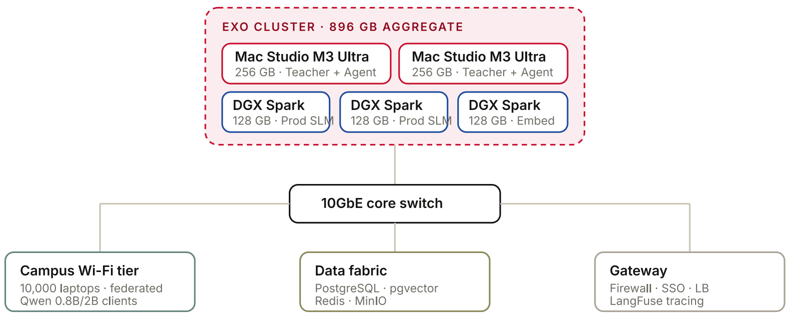


Figure — compute topology. EXO disaggregates model layers across nodes; no single node holds the full 35B at FP16.

896 GB
aggregate memory, 5 nodes

12U
wall cabinet — not a rack room

Desktop form factor
Two Mac Studios and three DGX Sparks on ventilated shelves. A 30-minute UPS rides through outages.

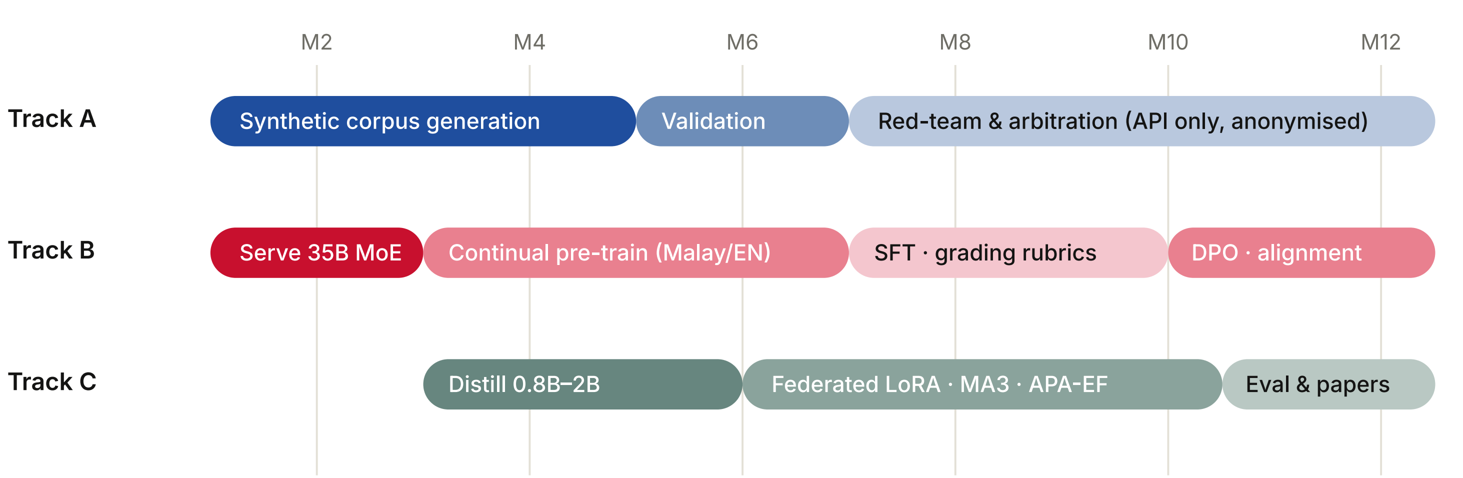
The privacy guarantee
The laptop swarm exchanges only LoRA adapter deltas with the cluster — **never raw student data**.

PART FIVE · THE PLAN

When.

Three tracks run in parallel across 12 months. The sovereign model is serving from Month 2 — value arrives early, not at the finish line.

Three tracks, each matched to what its hardware can verifiably do — from anonymised cloud teachers down to campus laptops.



A · Cloud teacher

Generates synthetic training data from anonymised rubrics. **No PII ever leaves UPSI.** ~RM 3–5K over 12 mo.

B · Sovereign teacher

The 35B MoE, served on-prem from Month 2 and continually adapted through six stages to DPO alignment.

C · Federated student

0.8–2B models distilled from the teacher, trained across 10,000 laptops under differential privacy.

SFT = supervised fine-tuning · DPO = direct preference optimisation · LoRA = low-rank adaptation

PART SIX · EXECUTION

How.

Four proofs turn the plan into a fundable reality: novel research, a provenance engine that makes AI show its receipts, governance enforced in code, and economics that break even.

The swarm is the research engine — three novel algorithms, evaluated at a 10,000-device scale few labs can replicate.

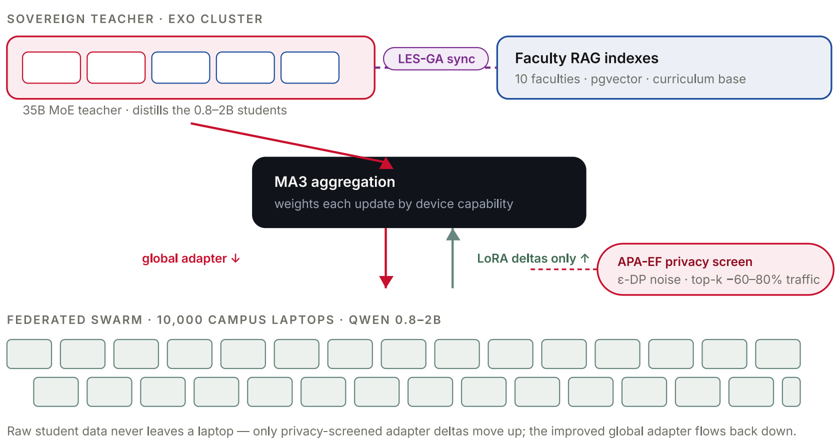


Figure — the swarm loop. Laptops train locally; APA-EF screens the updates; MA3 aggregates them into the teacher; LES-GA keeps faculty knowledge aligned.

MA3 AGGREGATION

Model-Agnostic Adaptive Aggregation

Weights adapter updates by device capability — Mac Studios and campus laptops reconciled in one round.

APA-EF PRIVACY

Adaptive Privacy-Aware Elastic Federated

ε-differential privacy with top-k sparsification — 60–80% less traffic, no individual update ever inspected.

LES-GA ALIGNMENT

Localized Elastic Search, Gradient-Aligned

Faculty-local RAG indexes stay consistent with the global curriculum base — no full-model retraining.

Foundational proof: Psyche

Nous Research's **Psyche** already proved transformers can be trained across untrusted internet nodes. SULUH operates under a **strictly easier trust assumption** — all 10,000 devices are institution-owned — which is exactly what enables APA-EF privacy and LES-GA alignment. Our contribution is deploying it **safely, privately, and educationally within a sovereign boundary**.

Chatbots fabricate up to half their citations. In education, that is disqualifying — IsRAG is the architecture built to end it.

55% / 18%
of citations fabricated by GPT-3.5 / GPT-4 in literature reviews (Walters & Wilder)

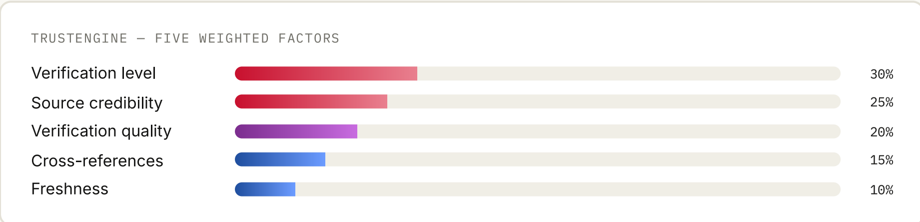
56%
of GPT-4o citations fake or erroneous in academic reviews (Deakin Univ., JMIR 2025)

58–88%
hallucination rates on legal queries across major models (Stanford)

Worst
where students need help most: niche, low-resource topics — exactly UPSI's Malay-language domain

EVERY ANSWER CARRIES ITS CHAIN

Provenance — every author, verifier & adapter, immutable → **Trust score** — 5-factor TrustEngine, 0–1 calibrated → **Cultural context** — 13-framework registry, incl. Malaysian & Islamic



A chatbot's fabricated citation looks identical to a real one — and a student cannot tell the difference. Retrieval alone doesn't fix it: even RAG systems still fabricate. **A citation tells you where; an isnad tells you whether to trust.** IsRAG verifies every claim against a provenance chain before it ever reaches a student.

Up to 10×
fewer false citations than today's chatbots — ≤5% golden-set target vs their 18–56%

100%
of answers auditable — IRB & PDPA-ready by design

Why this is the only viable answer for educational AI

A university cannot deploy a system that invents scholarship — every false citation is a defective teaching act with accreditation consequences. Frontier chatbots answer; **SULUH answers and proves**, adapting the Isnad (إسناد) — Islamic scholarship's millennium-old chain-of-transmission methodology — born naturally at a Malaysian university.

Sources: Walters & Wilder 2023 (42 topics) · Linardon et al., JMIR 2025 (GPT-4o: 19.9% fabricated + 45.4% erroneous) · Stanford HAI legal-hallucination studies · pooled 6-study fabrication rate 51% (PMC 2023) · Paper 1: IsRAG, target ACL/EMNLP/FAccT, evaluated on the 200-pair bilingual golden set

PDPA is enforced in code at one layer — not promised in policy — so no harness can bypass it, and every 2024 amendment maps to a gate.

GATE 01 **Consent & purpose**
Granular per-harness opt-in — `pdpa_gate.py` rejects non-consented queries at the API boundary.

GATE 02 **Data minimisation**
7-year retention with 3-year auto-purge, enforced by PostgreSQL TTL and a cron audit.

GATE 03 **Security**
AES-256 at rest, TLS 1.3 in transit, with automated certificate rotation.

GATE 04 **Federated privacy**
ε-differential privacy on every swarm update — no device identifier in the gradient payload.

GATE 05 **Accountability — the amendment's own checklist**
A registered DPO from Day 1 *(mandatory since Jun 2025)*, annual audit, 72-hour breach notification drill, IRB review, and full LangFuse trace logging — **the due-diligence evidence that shields UPSI's officers from personal liability**.

SULUH crosses below every cloud line by Month 14 — and saves RM 240–600K by Month 36.

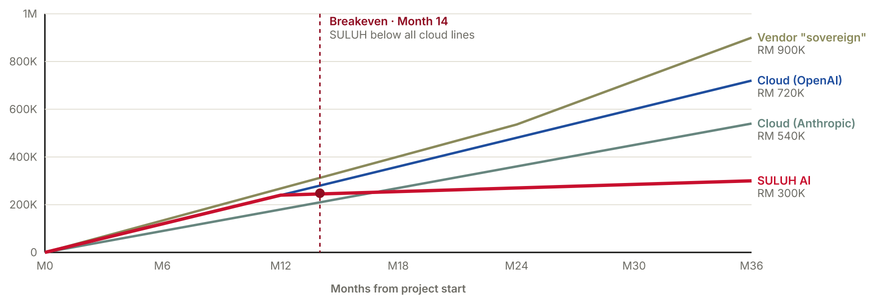


Figure — cumulative 36-month cost. SULUH's curve flattens after the build; cloud keeps accumulating at RM 15–25K/month.

Month 14
breakeven against every cloud option

RM 300K
36-month total cost of ownership

RM 30K
annual OPEX in steady state

And the asset is owned: capability, model weights, and data all stay with UPSI — plus the unquantified saving of **breach risk removed** (avg. Malaysian breach: RM 3.2M).

TCO = total cost of ownership · cloud lines assume RM 15–25K/month at UPSI-scale usage · breach cost: Simply Data Malaysia Threat Report 2025

Success is measured, not asserted — a bilingual golden-set evaluation and five planned publications.

TARGET METRICS · 200-PAIR GOLDEN SET

Grading accuracy (rubric adherence)	≥ 85%
Curriculum compliance (KPT/MQA)	≥ 90%
Malay language fluency	≥ 90%
Hallucination rate (IsRAG-verified)	≤ 5%
PDPA compliance	100%

Benchmarked against Kimi K3, Claude, Gemma 4 and GPT-4o · authored by 30+ UPSI researchers.

PLANNED PUBLICATIONS

1	**IsRAG** — Isnad-chain provenance RAG	ACL/EMNLP
2	**Dewan Council** — multi-agent deliberation	EMNLP
3	**SULUH AI** — sovereign 35B model	Nature MI
4	**Federated Swarm** — MA3, APA-EF, LES-GA	NeurIPS
5	**Deployment Study** — four-harness evaluation	LAK/AIED

Latency targets: <3s grading, <1s chat · IRB gate accuracy ≥98% · full metric set in the proposal appendix

We are asking with eyes open — every major risk has a named mitigation already in the plan.

RISK	IMPACT	MITIGATION IN PLAN
Model underperforms on bilingual tasks	High	Phase-0 benchmark of Qwen3.6 vs Gemma 4 vs Qwen3.5 before commitment; swap paths maintained
PDPA non-compliance	Critical	Privacy-by-design, DPO appointed Day 1, five governance gates in code, annual audit
SSO integration complexity	High	Phase-0 critical path, started Week 1 with Pusat ICT
Federated swarm attrition	Medium	Incentive design, gamification, academic credit for participating students
Phase-2 national compute never materialises	Medium	35B is already production-viable — 100B is upside, not a dependency

Full ten-item risk register with likelihood ratings in SULUH v3.3 §10

Approve RM 450K, and in 12 months UPSI owns Southeast Asia's first university-built sovereign educational AI.

Sovereign by design, provable by IsRAG, breakeven by Month 14, shielded from RM 1M liability by governance in code — with every byte of student data staying home.

THE INVESTMENT

RM 450K · 12 mo

200K hardware + 250K team, phased

THE RETURN

RM 240–600K saved

by Month 36 — plus breach risk removed

THE UPSIDE

5 papers · 4 systems

live at UPSI · first-mover seat · Phase-2 to 100B

"Pengetahuan Suluh Budiman" — the torch was always ours to carry. SULUH simply lights it for the age of AI.